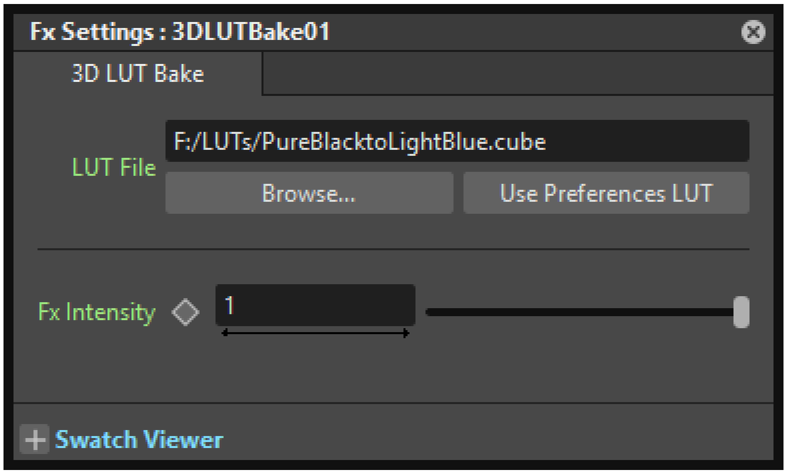
3D LUT Bake FX

Image Adjust / User guide

Apply a selected 3D LUT to rendered image colors. Connect the image or composition to **Source**. For an overall look, place this node after the completed composition. Alpha is preserved.



Earlier dialog appearance. The current copy button is labeled **Copy Preferences LUT** and appears separately from the file field.

Controls and accepted entries

LUT File - Enter an existing, readable **.cube** or supported **.3dl** file, using a full absolute path such as `F:/LUTs/Dusty_Orange.3dl`. Do not add quotation marks or enter a web URL. A blank path bypasses the effect. The path is not animated and the LUT file is not embedded in the scene.

Browse... - Select a LUT from the filtered file picker. Cancel leaves the path unchanged. Typed, browsed, and copied paths are validated before assignment. An invalid selection keeps the previous path and displays an error below the controls.

Copy Preferences LUT - Copies the path configured for the current monitor, even when display calibration is disabled. This is a one-time copy: later monitor or Preferences changes do not update the FX. The file must remain available at the copied path.

Fx Intensity - Enter a number from **0 to 1**, or use the slider. The default **1** applies the full effect; **0** removes its contribution. Intermediate values blend the effect. Use the diamond control to animate intensity. Fix an invalid file path instead of relying on intensity to suppress errors.

Swatch Viewer (+) - Expand the local FX preview. This viewing control accepts no text or numeric entry. Use the **Help** icon in FX Settings to open this guide.

Supported files and results

File compatibility, color handling, and recovery

Accepted LUT data

.cube - A 3D table with `LUT_3D_SIZE` from **2 to 129**, followed by exactly size cubed RGB triples of finite floating-point numbers. `TITLE` and comment lines are accepted. `DOMAIN_MIN` and `DOMAIN_MAX` specify three input bounds; alternatively, `LUT_3D_INPUT_RANGE` specifies one shared minimum and maximum. Do not combine these forms. Each minimum must be below its maximum. The default input range is **0 to 1**. 1D, 2D, and shaper LUTs are unsupported.

.3dl - Only **Lustre-format** files with a `3DMESH` header are supported. Headerless **Flame-format** files are unsupported. The next line must be `Mesh n b`: **n** is the grid exponent (**0 to 7**) and **b** is the output bit depth (**1 to 30**). Each axis contains $\mathbf{2^n + 1}$ samples, followed by that grid size cubed integer RGB triples.

The input grid must increase strictly, span the full **8-, 10-, 12-, or 16-bit** range (0 to 255, 1023, 4095, or 65535), and be uniformly spaced, allowing rounding differences of at most one code value. Nonuniform grids are rejected. Integer RGB values are normalized using $\mathbf{2^b - 1}$; the FX clips transformed output to **0 to 1**. Extra color entries are rejected; the optional supported footer is `LUT8` with `gamma 1.0`.

Display calibration and baked output

This FX changes **rendered pixels**. Preferences display calibration changes their **appearance on the current monitor**. A monitor-calibration LUT compensates for a particular display; baking that correction does not guarantee the same appearance on another monitor.

When the same LUT is used in both places, the displayed baked result receives the transform again. Account for that extra display application when comparing the two. Do not generally disable monitor calibration to evaluate a separate creative look LUT.

The effect evaluates the LUT in nonlinear RGB; the FX framework handles surrounding linear-color conversions. Baking does not inherently leave the linear-rendering workflow. Compare Preview and saved output with equivalent render settings, and distinguish saved pixel values from displayed appearance.

Limits and troubleshooting

Inputs are clamped to the LUT domain. Transformed RGB is clipped to **0 to 1**, including floating-point output, so this is not an unrestricted HDR transform. Transparent edges are processed with alpha taken into account.

If a selected file is later missing or malformed, rendering stops at the first computation error and reports the failure once; an incomplete result is not opened automatically. No identity LUT or previously loaded LUT is silently substituted. Repair the file or choose a valid path, then retry. Keep external LUT files with the scene when moving it to another computer.